L6. Statistics

Lying with statistics

What is it?

> Lying with statistics means **using correct statistical tools in the wrong way on purpose**.
>
> The data may be real, and the method may be from a standard statistics book, but the person **chooses the wrong method or wrong presentation** in order to make others believe something false.
>
> The key idea is *intention*: it is only lying if the person **knows** the method is misleading and still uses it.

Example(s):

- Using the **mean** income instead of the **median** to make a family seem richer or poorer, even though the median is the better measure in that situation.
- Showing a graph **without a zero point** on the y-axis to make a small change look big.
- Saying a medical treatment has a **"33% risk reduction"** instead of saying **"1.3 percentage point reduction"**, to make the result look more impressive.
- Reporting that a new medicine "**doubles the risk of blood clots**" without showing that the increase is from **1 in 7000 to 2 in 7000**, which changes how dangerous it really seems.
- Choosing a statistical test that gives the **desired result** instead of the correct one for the situation.

Statistical methodology

What is it?

Statistical methodology is about **explaining why you chose a specific statistical method** for your data and your purpose.

It is not enough to just run a test. You must **justify** why this test is the right one in this situation.

Good methodology helps prevent mistakes and avoids misleading people.

Examples:

- Choosing the **mean** instead of the **median**, and explaining: "I use the mean because I care about total wealth, not inequality."
- Explaining why you use a **t-test** instead of a **chi-square test**, based on the type of data you have.
- Before running a significance test, stating **why** you chose a particular significance level (like 0.05).
- When showing a graph, explaining why you **kept the zero point** to avoid misleading the viewer.
- Choosing **Bayesian methods** because you want to measure *degrees of belief*, and explaining why this fits your goal better than simply rejecting a hypothesis.

Descriptive statistics

What is it?

Descriptive statistics is about **showing data clearly and accurately**.

It uses tools like averages, graphs, tables, and percentages to **summarize** what the data looks like without trying to make deeper predictions.

The goal is to present information in a way that is **honest, clear, and easy to understand**.

Example(s):

- Using the **median** household wealth to show the "middle" person's wealth in a country.

- Making a **bar chart** that starts at zero so the differences are not exaggerated.
- Reporting the **mean temperature** for the year.
- Showing how many people fall into each age group using a **histogram**.
- Using the **absolute risk reduction** (from 3.9% to 2.6%) to clearly communicate how much a medical treatment changes risk.

Inferential statistics

What is it?

Inferential statistics is about **using data to make justified conclusions** about something bigger than the data itself.

You use a sample to make statements about a population, test hypotheses, or estimate effects.

The goal is not just to describe the data, but to **reason beyond it** in a careful and justified way.

Example(s):

- Testing whether a new medicine works by comparing two groups and using a **significance test**.
- Using a **confidence interval** from a sample survey to estimate the true average income of a country.
- Running a **hypothesis test** to see whether a coin is fair.
- Using **Bayesian methods** to update your belief about a hypothesis after seeing new data.
- Determining if a change (e.g., 75% improved after treatment) is **statistically significant** based on sample size and probability.

Stochastic hypothesis

What is it?

A stochastic hypothesis is a hypothesis that **does not give a certain prediction**, but instead says that outcomes happen with **certain probabilities**.

It tells you *how likely* different results are, not which result must happen.

Example(s):

- **Mendel's genetics:**

  When crossing certain pea plants, the hypothesis predicts a **50% chance** of yellow peas and a **50% chance** of green peas—not a guaranteed outcome.

- **Rolling a die:**

  The hypothesis "each side has a 1/6 chance" is stochastic because it states probabilities, not certainties.

- **Weather forecasts:**

  "There is a 40% chance of rain tomorrow" is a stochastic claim.

- **Medical test accuracy:**

  A test might detect a disease with **95% probability**, not with certainty.

Deterministic hypothesis

What is it?

A deterministic hypothesis is a hypothesis that makes **certain**, not probabilistic, predictions.

If the hypothesis is true, then the outcome **must** happen exactly as it says—there is no randomness involved.

Example(s):

- **Chemistry reaction law:**

  “If you heat this substance to 100°C at sea level, it will boil.”

  The prediction is certain given the conditions.

- **Physics (classical mechanics):**

  “If you drop an object in a vacuum, it will fall with constant acceleration.”

  No randomness is involved.

- **Medical example from the lecture:**

  “If a person has Alzheimer’s disease, they will have amyloid plaque in the brain.”

  According to the hypothesis, this must be true for every case.

- **Mathematics:**

  “If a number is even, dividing it by 2 gives an integer.”

  Always true, no exceptions.

Quantitative measure of measurement error

What is it?

A quantitative measure of measurement error is a way to express **how likely it is that an observation is wrong**, using **numbers**.

Instead of simply saying “there might be an error,” we give a **numerical probability** that a mistake in measurement has occurred.

This helps scientists decide **how strongly** an observation should count against a hypothesis.

Example(s):

- A medical test might miss a disease **5% of the time**.

  → The measurement error probability is **0.05**.

- A microscope study might fail to detect amyloid plaque **1 out of 50 times**.

  → Error rate = **0.02**.

- A thermometer may have an accuracy range such that readings are wrong **in 3% of cases**.

  → Error probability = **0.03**.

- In quality control, a machine might incorrectly classify products **2 times per 100 measurements**.

  → Error probability = **0.02**.

Error-based statistics

What is it?

Error-based statistics is about calculating **how probable an observation is if the hypothesis were true**.

Instead of asking "Is the hypothesis true?", we ask:

"If the hypothesis *is* true, how likely is it to see this data?"

If the observation would be **very unlikely** under the hypothesis, then the hypothesis should be **rejected**.

Example(s):

- **Coin fairness test:**

  Hypothesis: "The coin is fair."

  Observation: Only **4 heads in 20 flips**.

  Error-based statistics asks:

  How likely is it to get 4 heads if the coin is fair?

  → If the probability is extremely low, we reject the hypothesis.

- **Genetics experiment:**

  Hypothesis: Offspring traits follow a **50/50** distribution.

  If we observe **90%** of one type, we calculate how probable that is **assuming** the 50/50 model is true.

- **Medical test accuracy:**

  Hypothesis: "This patient does **not** have the disease."

  We ask:

  What is the chance of getting this test result if the patient is disease-free?

  If the chance is very low, we question the hypothesis.

- **Alzheimer's example from the lecture:**

  Hypothesis: "All Alzheimer's patients have plaque."

  If we observe a patient without plaque, we ask:

  How likely is a measurement error?

  If the chance of error is small, the hypothesis is rejected.

Fisher's significance testing

What is it?

Fisher's significance testing is a method for testing a hypothesis by asking:

"If the hypothesis were true, how likely is it to observe data this extreme?"

It uses the **p-value** (a probability) to decide whether the data is too unlikely under the hypothesis.

If the p-value is **smaller** than a chosen significance level (often 0.05), the hypothesis is **rejected**.

The method tests **one hypothesis** at a time and does *not* claim the hypothesis is true—only that it is not rejected.

Example(s):

- **Coin fairness test:**

  Hypothesis: "The coin is fair."

  Observation: 4 heads out of 20 flips.

  Compute p-value = probability of getting 4 or fewer heads if the coin is fair.

  If this probability is very small (like 0.001), we reject the hypothesis.

- **Medicine effect test:**

  Hypothesis: “The medicine has no effect.”

  If the observed improvement would be extremely unlikely under “no effect,” the p-value is low, and the hypothesis is rejected.

- **Classroom experiment:**

  Hypothesis: “Students score equally well before and after training.”

  If most students improve in a way unlikely to happen by chance, the test rejects the hypothesis.

Test statistic

What is it?

A test statistic is **a number you calculate from your sample** that helps decide whether to reject a hypothesis.

It reduces the data to one meaningful value.

Example(s):

- Number of **heads** in 20 coin flips.
- The **average** improvement before/after a treatment.
- The **difference** between two group means.

Sampling distribution

What is it?

A sampling distribution is the **distribution of all possible values** the test statistic can take **if the hypothesis is true**.

It tells you what outcomes are **likely** or **unlikely** under the hypothesis.

Example(s):

- For a fair coin, the sampling distribution gives the probability of getting

0, 1, 2, ..., 20 heads in 20 flips.

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p-value

What is it?

The p-value is **the probability of getting the observed test statistic, or something even more extreme**, **if the hypothesis is true**.

It measures how surprising your data is under the hypothesis.

Example(s):

- If you get 4 heads in 20 flips, the p-value is the probability of getting **4 or fewer heads** assuming a fair coin.

Significance level

What is it?

The significance level (often **0.05**) is a **threshold** used to decide whether to reject a hypothesis.

If the p-value is **below** this level, the hypothesis is **rejected**.

Example(s):

- p-value = 0.001 < 0.05 → Reject the hypothesis.
- p-value = 0.12 > 0.05 → Do not reject the hypothesis.

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p-value abuse

What is it?

p-value abuse happens when someone **intentionally changes the way a test is done**—such as the method, the sample, or the way the statistic is calculated—in order to get a p-value that gives the result they *want*.

This is dishonest because the p-value should reflect the actual test, not a manipulated one.

Examples:

- **Repeating the experiment** until one trial gives a “significant” p-value and reporting only that trial.
- **Changing the way the test statistic is counted**, for example grouping outcomes differently to get a lower p-value.
- **Dropping certain data points** that make the result less significant.
- **Switching to a different statistical test** that gives a more favourable p-value.
- **Lowering the significance level** (e.g., from 0.01 to 0.05) to avoid rejecting the hypothesis.

Neyman–Pearson hypothesis testing

What is it?

Neyman–Pearson hypothesis testing is a method for comparing **two opposite hypotheses** and deciding which one to accept.

It focuses on **controlling error rates** (type I and type II errors) and designing a test with enough **power** to make good decisions.

You must always set up **two hypotheses**:

Original hypothesis (Hi)

What is it?

This is the main claim you want to test.

It is treated as the hypothesis that might be rejected.

Example:

Hi: *The coin is fair.*

Alternative hypothesis (Ha)

What is it?

Ha is the **logical opposite** of Hi.

Only one of the two can be true.

If Hi is false, Ha **must** be true.

Example:

Ha: *The coin is not fair.*

Type I error

What is it?

A type I error happens when you **wrongly reject a true Hi**.

Example:

The coin *is* fair, but your test mistakenly concludes it is not.

("False positive.")

Type II error

What is it?

A type II error happens when you **wrongly accept a false Hi**.

Example:

The coin is actually biased, but your test concludes it is fair.

("False negative.")

Power of a test

What is it?

The power of a test is the **probability that the test correctly rejects a false original hypothesis (Hi)**.

In other words, if Hi is actually false, a high-power test will usually detect this and point you toward the alternative hypothesis (Ha).

A test with **low power** often fails to notice real effects.

A test with **high power** is good at detecting when something is wrong with Hi.

Example(s):

- If a medical test has **high power**, it will correctly identify that a disease is present when it really is present.
- In a coin experiment, if the coin is biased but your test only has low power, you might **fail to detect** this bias.
- Increasing the **sample size** (more data) usually increases the power of the test, making results more reliable.

Bayesian statistics

What is it?

Bayesian statistics is a way of evaluating hypotheses by **updating your belief** in a hypothesis after seeing new data.

You start with a **prior probability** (your initial belief), look at the **observed outcome**, and then use **Bayes' theorem** to calculate the **posterior probability**, which is your new, updated belief.

This method treats probability as a **degree of belief**, not just a frequency in data.

Example(s):

- You think there is a **60% chance** a coin is fair (prior).

After flipping it 20 times and getting only 4 heads, you update your belief and maybe conclude there's only a **3% chance** it is fair (posterior).

- Before testing a medicine, you may think it probably works (**prior = 0.7**).

  After seeing strong positive results, your belief increases (**posterior = 0.9**).

- A doctor starts with a low belief a patient has a disease (**prior**).

  After a positive test result, the belief increases according to Bayes' theorem (**posterior**).

The problem of old evidence

What is it?

The problem of old evidence is a difficulty in Bayesian statistics:

It is hard to decide **whether evidence that has already been used to update your beliefs should be used again**.

Bayes' theorem assumes that new data should update your prior beliefs only **once**.

But in real life, people often re-read or re-encounter old evidence, and the theory does not clearly say how to handle this.

Example(s):

- You hear about a scientific experiment and update your belief in a hypothesis.

  Weeks later, you read about the same experiment again and feel like your confidence increases even more — but according to Bayesian rules, you **should not use the same evidence twice**.

- A physicist already believed a theory because of certain experiments ("old evidence").

  When the theory is later formalized, the same old evidence cannot be used again to raise the posterior probability, even though it strongly supports the theory.

- If you tell a friend about your experiment today, and they update their belief,

but then they read your published paper next month and update again,

this “double counting” violates Bayesian principles — but the theory does not clearly say how to prevent it.

The problem of uncertain evidence

What is it?

The problem of uncertain evidence is that Bayesian statistics usually assumes that **all evidence is absolutely correct** (probability = 1).

But in real science, evidence is often **uncertain**, noisy, or possibly wrong.

Classic Bayesian updating does not have a clear way to handle cases where the evidence itself has a probability **less than 1** of being correct.

In other words, Bayes' theorem expects you to update beliefs based on evidence you fully trust but in reality, many observations are **fallible**.

Example(s):

- A medical test might be **95% accurate**, not 100%.

  Bayesian updating says you should update based on “The test is positive,” treating it as certain, even though the test could be wrong.

- A brain scan for amyloid plaque might miss plaque in **some cases**.

  But Bayesianism would treat “Plaque was not observed” as a certain fact, even though it may be an error.

- In physics or biology, measurements can have **random noise** or **equipment error**.

  Standard Bayesian updating does not cleanly incorporate the chance that the measurement is incorrect.

- If a witness reports an event but is only **70% reliable**, Bayesianism does not directly say how to use that partially trustworthy evidence to update beliefs.

The problem of priors

What is it?

The problem of priors is that Bayesianism requires you to **choose a prior probability**—your initial belief in a hypothesis before seeing data—but it does **not give a clear rule** for how this choice should be made.

Different people may choose very different priors, and this can lead to very different posterior probabilities, even with the same data.

The principle principle

What is it?

The principle principle says that if you **know the objective (real-world) probability** of an outcome, then your **prior** should match that probability.

In other words, your belief should follow the actual known chance.

Example:

If a fair die has a 1/6 chance of landing on a 4, your prior for "the next roll is a 4" should be **1/6**, not a random number like 0.3.

The principle of indifference

What is it?

The principle of indifference says that if you **have no information at all** about the chances of different outcomes, then your prior probabilities should be **split equally** among all possibilities.

Example(s):

- If a coin is unfamiliar and you know nothing about it, assign

  P(heads) = 0.5, P(tails) = 0.5

- If a box might contain 3, 4, or 5 balls but you have no clue which, assign

  P(3 balls) = P(4 balls) = P(5 balls) = 1/3
